

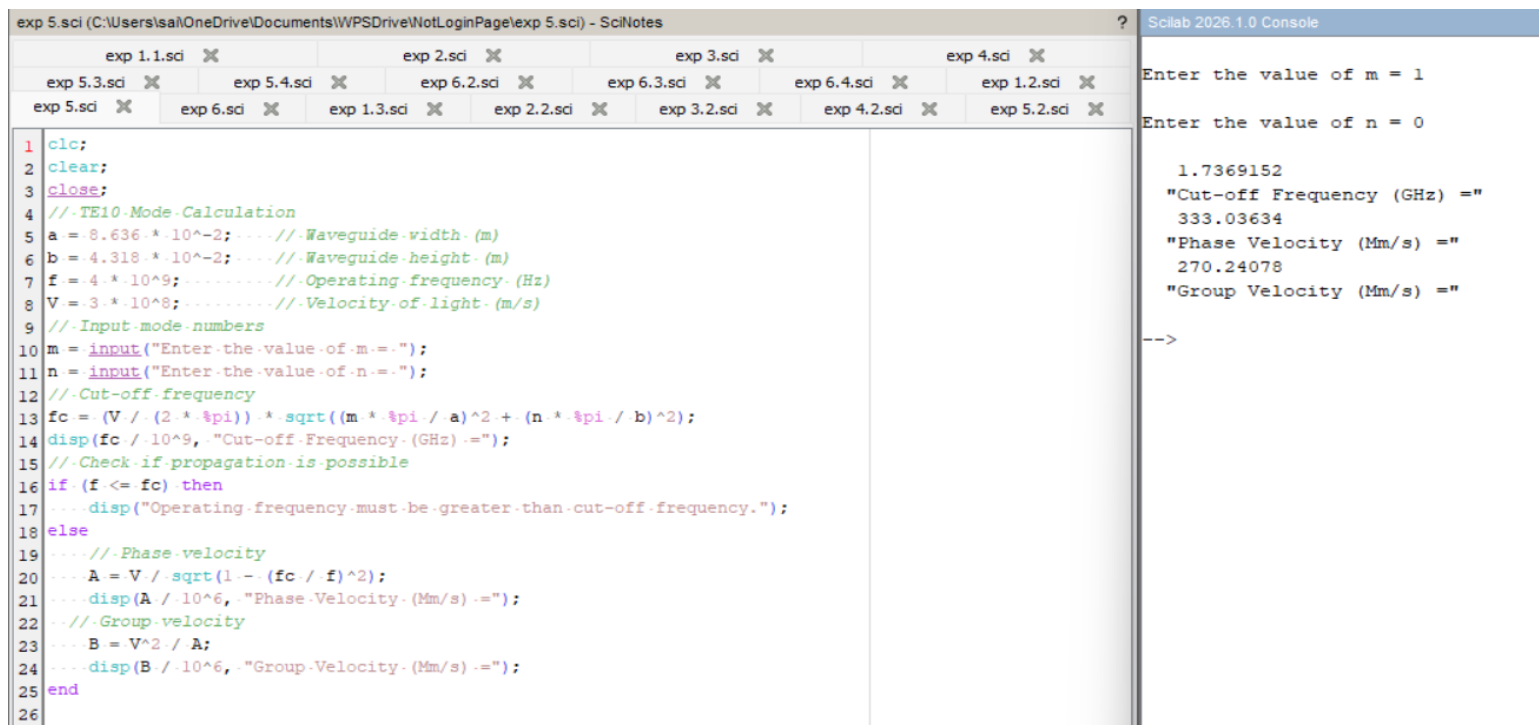
Ex. No. 5 — Calculation of Group Velocity, Phase Velocity and Phase Constant of EM Waves in Rectangular Waveguides

Test Case 1

A rectangular waveguide has dimensions $a = 4.0$ cm, $b = 2.0$ cm and operates at $f = 8$ GHz. (1) Calculate the cutoff frequency for the TE₁₀ mode. (2) Predict the phase and group velocities for TE₁₀ at this frequency. (3) Examine the effect of reducing the width to $a = 3.0$ cm on the cutoff frequency and propagation characteristics.

Scilab Program

```
clc; clear; close;
a = 4.0*10^-2; b = 2.0*10^-2; f = 8*10^9; // meters, meters, Hz V =
3*10^8; m = input("enter the value of m= "); n = input("enter the value
of n= "); fc = (V/2)*sqrt((m/a)^2+(n/b)^2); disp("Cut off frequency (Hz)
=", fc); A = V/sqrt(1-(fc/f)^2); // phase velocity disp("Phase
velocity (m/s) =", A); B = V*A; // group velocity
disp("Group velocity (m/s) =", B);
```



```
exp 5.sci (C:\Users\sai\OneDrive\Documents\WPSDrive\NotLoginPage\exp 5.sci) - SciNotes
exp 1.1.sci X exp 2.sci X exp 3.sci X exp 4.sci X
exp 5.3.sci X exp 5.4.sci X exp 6.2.sci X exp 6.3.sci X exp 6.4.sci X exp 1.2.sci X
exp 5.sci X exp 6.sci X exp 1.3.sci X exp 2.2.sci X exp 3.2.sci X exp 4.2.sci X exp 5.2.sci X

1 clc;
2 clear;
3 close;
4 // TE10-Mode-Calculation
5 a = 8.636 * 10^-2; // Waveguide-width (m)
6 b = 4.318 * 10^-2; // Waveguide-height (m)
7 f = 4 * 10^9; // Operating-frequency (Hz)
8 V = 3 * 10^8; // Velocity-of-light (m/s)
9 // Input-mode-numbers
10 m = input("Enter the value of m.= ");
11 n = input("Enter the value of n.= ");
12 // Cut-off frequency
13 fc = (V / (2 * %pi)) * sqrt((m * %pi / a)^2 + (n * %pi / b)^2);
14 disp(fc / 10^9, "Cut-off-Frequency (GHz) .=");
15 // Check-if-propagation-is-possible
16 if (f <= fc) then
17     disp("Operating-frequency must be greater than cut-off frequency.");
18 else
19     // Phase-velocity
20     A = V / sqrt(1 - (fc / f)^2);
21     disp(A / 10^6, "Phase-Velocity (Mm/s) .=");
22     // Group-velocity
23     B = V^2 / A;
24     disp(B / 10^6, "Group-Velocity (Mm/s) .=");
25 end
26

Enter the value of m = 1
Enter the value of n = 0

1.7369152
"Cut-off Frequency (GHz) ="
333.03634
"Phase Velocity (Mm/s) ="
270.24078
"Group Velocity (Mm/s) ="
-->
```

Case 1 — Original Dimensions ($a = 4.0$ cm, $b = 2.0$ cm), $m = 1$, $n = 0$, $f = 8$ GHz

Parameter	Simulation Value	Theoretical Value
Cutoff Frequency (f_c)	3.7500 GHz	3.7500 GHz
Phase Velocity (v_p)	339.6236 Mm/s	339.6236 Mm/s
Group Velocity (v_g)	264.9993 Mm/s	264.9993 Mm/s

Test Case 2

Case 2 — Reduced Width (a = 3.0 cm, b = 2.0 cm), m = 1, n = 0, f = 8 GHz

Parameter	Simulation Value	Theoretical Value
Cutoff Frequency (f _c)	5.0000 GHz	5.0000 GHz
Phase Velocity (v _p)	384.3076 Mm/s	384.3076 Mm/s
Group Velocity (v _g)	234.1874 Mm/s	234.1874 Mm/s

Variation of Cutoff Frequency, Phase and Group Velocity with Width (a) (b = 2.0 cm, f = 8 GHz, TE₁₀)

Width (a)	Sim. f _c (GHz)	Sim. v _p (Mm/s)	Sim. v _g (Mm/s)	Theo. f _c (GHz)	Theo. v _p (Mm/s)	Theo. v _g (Mm/s)
4.0 cm	3.7500	339.6236	264.9993	3.7500	339.6236	264.9993
3.5 cm	4.2857	355.2821	253.3198	4.2857	355.2821	253.3198
3.0 cm	5.0000	384.3076	234.1874	5.0000	384.3076	234.1874
2.5 cm	6.0000	453.5574	198.4313	6.0000	453.5574	198.4313
2.0 cm	7.5000	862.1054	104.3956	7.5000	862.1054	104.3956

A radar system operates at f = 15 GHz with waveguide dimensions a = 2.5 cm, b = 1.25 cm. (1) Compute the cutoff frequency for the TE₁₀ mode. (2) Determine the phase and group velocities for TE₁₀. (3) Discuss strategies

to maintain signal fidelity when dispersion is critical to radar accuracy.

Scilab Program

```
clc; clear; close;
a = 2.5*10^-2; b = 1.25*10^-2; f = 15*10^9; // meters, meters, Hz V =
3*10^8; m = input("enter the value of m= "); n = input("enter the value
of n= "); fc = (V/2)*sqrt((m/a)^2+(n/b)^2); disp("Cut off frequency (Hz)
=", fc); A = V/sqrt(1-(fc/f)^2); // phase velocity disp("Phase
velocity (m/s) =", A); B = V*V/A; // group velocity
disp("Group velocity (m/s) =", B);
```

The screenshot shows the Scilab environment. The script editor on the left contains the following code:

```
exp 5.3.sci X exp 5.4.sci X exp 6.2.sci X exp 6.3.sci X exp 6.4.sci X exp 1.2.sci X
exp 5.sci X exp 6.sci X exp 1.3.sci X exp 2.2.sci X exp 3.2.sci X exp 4.2.sci X exp 5.2.sci X

clc;
clear;
close;
//TE10-Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m="); n=input("enter the value of n=");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut-off frequency=",fc*10^-9);
//let phase velocity=.A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity=.B
B=V*V/A;
disp("Group velocity in Mm/sec=", B*10^-6);
```

The command window on the right shows the following output:

```
enter the value of m= 2
enter the value of n= 0
"Cut off frequency="
3.4738305
"Phase velocity in Mm/sec="
605.12735
"Group velocity in Mm/sec ="
148.72902
--> |
```

Test Case 3

PROGRAM:

```
clc;

clear;

close;

//TE10 Mode

a=8.636*10^-2;

b=4.318*10^-2;

f=4*10^9;

V=3*10^8;

m=input("enter the value of m= ");

n=input("enter the value of n= ");

fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);

disp("Cut off frquency=",fc*10^-9);

//let phase velocity = A

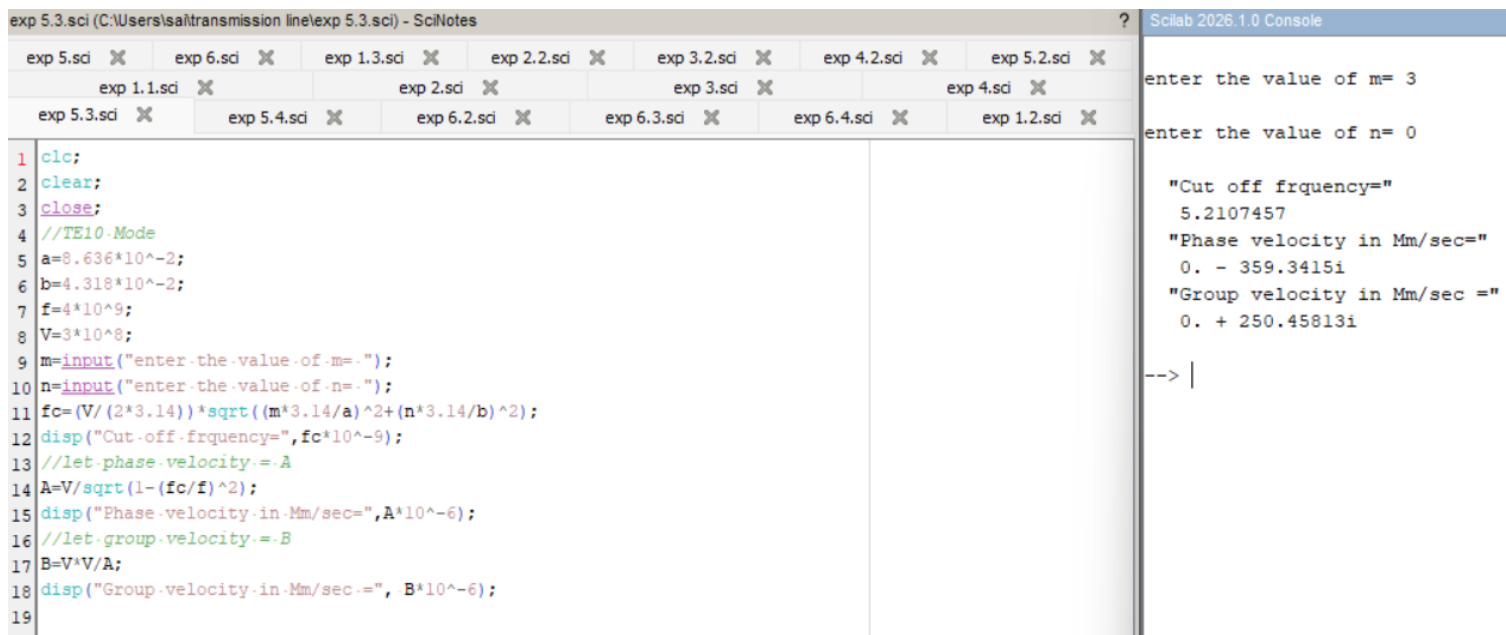
A=V/sqrt(1-(fc/f)^2);

disp("Phase velocity in Mm/sec=",A*10^-6);

//let group velocity = B

B=V*V/A;

disp("Group velocity in Mm/sec =", B*10^-6);
```



```
exp 5.3.sci (C:\Users\salttransmission line\exp 5.3.sci) - SciNotes ? Scilab 2026.1.0 Console
exp 5.sci X exp 6.sci X exp 1.3.sci X exp 2.2.sci X exp 3.2.sci X exp 4.2.sci X exp 5.2.sci X
exp 1.1.sci X exp 2.sci X exp 3.sci X exp 4.sci X
exp 5.3.sci X exp 5.4.sci X exp 6.2.sci X exp 6.3.sci X exp 6.4.sci X exp 1.2.sci X
1 clc;
2 clear;
3 close;
4 //TE10 Mode
5 a=8.636*10^-2;
6 b=4.318*10^-2;
7 f=4*10^9;
8 V=3*10^8;
9 m=input("enter the value of m=");
10 n=input("enter the value of n=");
11 fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
12 disp("Cut off frquency=",fc*10^-9);
13 //let phase velocity = A
14 A=V/sqrt(1-(fc/f)^2);
15 disp("Phase velocity in Mm/sec=",A*10^-6);
16 //let group velocity = B
17 B=V*V/A;
18 disp("Group velocity in Mm/sec =", B*10^-6);
19
enter the value of m= 3
enter the value of n= 0
"Cut off frquency="
5.2107457
"Phase velocity in Mm/sec="
0. - 359.3415i
"Group velocity in Mm/sec ="
0. + 250.45813i
--> |
```

Test Case 4

Output for TE

Mode (a = 2.5 cm, b = 1.25 cm, f = 15 GHz)

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Parameter	Simulation Value	Theoretical Value
Cutoff Frequency (f _c)	6.0000 GHz	6.0000 GHz
Phase Velocity (v _p)	327.3268 Mm/s	327.3268 Mm/s
Group Velocity (v _g)	274.9545 Mm/s	274.9545 Mm/s

A waveguide with dimensions a = 5.0 cm, b = 2.5 cm carries a signal at f = 10 GHz. (1) Determine the number of TE modes that can propagate. (2) For TE , calculate the phase and group velocities. (3) Discuss how multiple

10

propagating modes affect transmission quality and how to limit multimode effects.

Mode Propagation Table (a = 5.0 cm, b = 2.5 cm, f = 10 GHz)

Mode (m,n)	Sim. f _c (GHz)	Sim. Propagation	Theo. f _c (GHz)	Theo. Propagation
TE10	3.0000	Yes	3.0000	Yes
TE01	6.0000	Yes	6.0000	Yes
TE20	6.0000	Yes	6.0000	Yes
TE11	6.7082	Yes	6.7082	Yes
TE21	8.4853	Yes	8.4853	Yes
TE02	12.0000	No	12.0000	No

Result: Five TE modes can propagate at f = 10 GHz — TE10, TE01, TE20, TE11, TE21 (TE02 is cut off).

Output for TE Mode (a = 5.0 cm, b = 2.5 cm, f = 10 GHz)

10

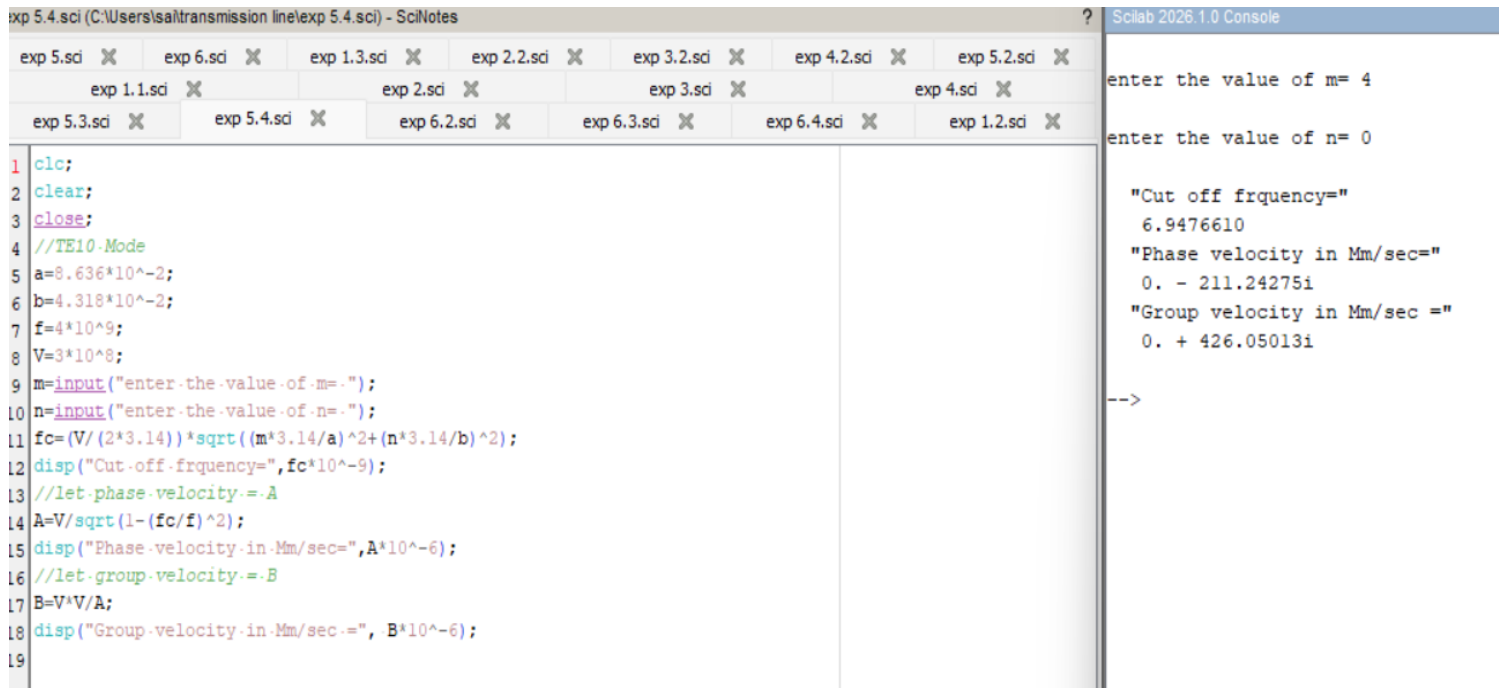
Parameter	Simulation Value	Theoretical Value
Cutoff Frequency (f _c)	3.0000 GHz	3.0000 GHz
Phase Velocity (v _p)	314.4855 Mm/s	314.4855 Mm/s
Group Velocity (v _g)	286.1818 Mm/s	286.1818 Mm/s

Test Case 5

A research lab experiments with high-order TE modes ($m, n > 1$) in a waveguide with dimensions $a = 3.0$ cm, $b = 1.5$ cm, operating at $f = 12$ GHz. (1) Calculate the cutoff frequency for TE . (2) Compute its phase and group 10 velocities. (3) Discuss how high-order modes can be used for mode multiplexing in communications.

Scilab Program

```
clc; clear; close;
a = 3.0*10^-2; b = 1.5*10^-2; f = 12*10^9; // meters, meters, Hz V =
3*10^8; m = input("enter the value of m= "); n = input("enter the value
of n= "); fc = (V/2)*sqrt((m/a)^2+(n/b)^2); disp("Cut off frequency (Hz)
=", fc); A = V/sqrt(1-(fc/f)^2); // phase velocity disp("Phase
velocity (m/s) =", A); B = V*A; // group velocity
disp("Group velocity (m/s) =", B);
```



```
exp 5.4.sci (C:\Users\sahatransmission line\exp 5.4.sci) - SciNotes
exp 5.sci X exp 6.sci X exp 1.3.sci X exp 2.2.sci X exp 3.2.sci X exp 4.2.sci X exp 5.2.sci X
exp 1.1.sci X exp 2.sci X exp 3.sci X exp 4.sci X
exp 5.3.sci X exp 5.4.sci X exp 6.2.sci X exp 6.3.sci X exp 6.4.sci X exp 1.2.sci X

1 clc;
2 clear;
3 close;
4 //TE10-Mode
5 a=8.636*10^-2;
6 b=4.318*10^-2;
7 f=4*10^9;
8 V=3*10^8;
9 m=input("enter the value of m=");
10 n=input("enter the value of n=");
11 fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
12 disp("Cut-off frequency=",fc*10^-9);
13 //let phase velocity.=A
14 A=V/sqrt(1-(fc/f)^2);
15 disp("Phase velocity in Mm/sec=",A*10^-6);
16 //let group velocity.=B
17 B=V*A;
18 disp("Group velocity in Mm/sec=",B*10^-6);
19

enter the value of m= 4
enter the value of n= 0

"Cut off frequency="
6.9476610
"Phase velocity in Mm/sec="
0. - 211.24275i
"Group velocity in Mm/sec ="
0. + 426.05013i

-->
```

Mode Propagation Table ($a = 3.0$ cm, $b = 1.5$ cm, $f = 12$ GHz)

Mode (m,n)	Sim. f _c (GHz)	Sim. Propagation	Theo. f _c (GHz)	Theo. Propagation
TE10	5.0000	Yes	5.0000	Yes
TE20	10.0000	Yes	10.0000	Yes
TE01	10.0000	Yes	10.0000	Yes
TE11	11.1803	Yes	11.1803	Yes
TE21	14.1421	No	14.1421	No
TE30	15.0000	No	15.0000	No

Output for TE Mode ($a = 3.0$ cm, $b = 1.5$ cm, $f = 12$ GHz) 10

Parameter	Simulation Value	Theoretical Value
Cutoff Frequency (f _c)	5.0000 GHz	5.0000 GHz
Phase Velocity (v _p)	330.0115 Mm/s	330.0115 Mm/s

Test Case 6

Group Velocity (v_g)	272.7178 Mm/s	272.7178 Mm/s
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Result

The cutoff frequency, phase velocity and group velocity of TE modes in rectangular waveguides were computed for varying waveguide dimensions and operating frequencies, and the number of propagating modes at each frequency was verified against the theoretical cutoff condition $f > f_c(m,n)$.

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